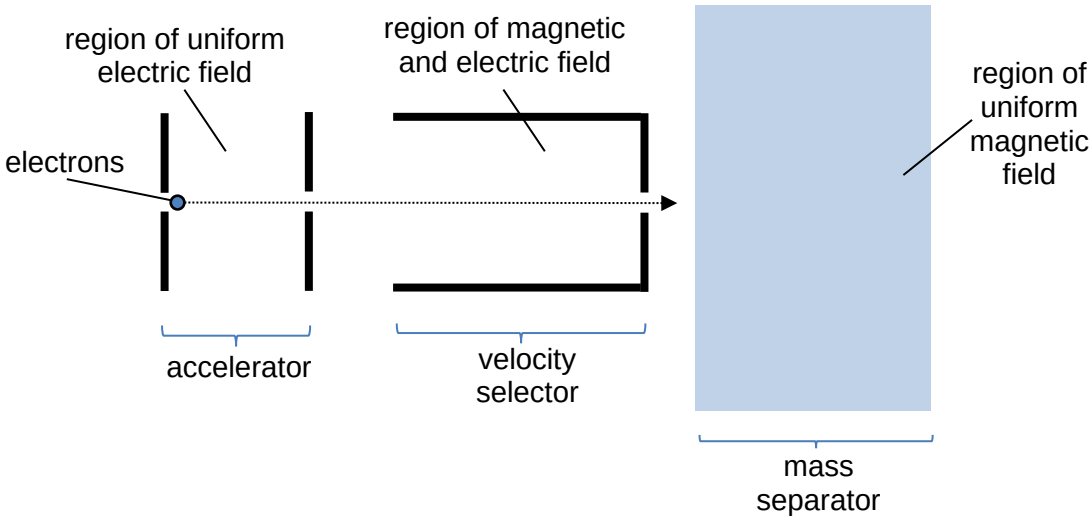
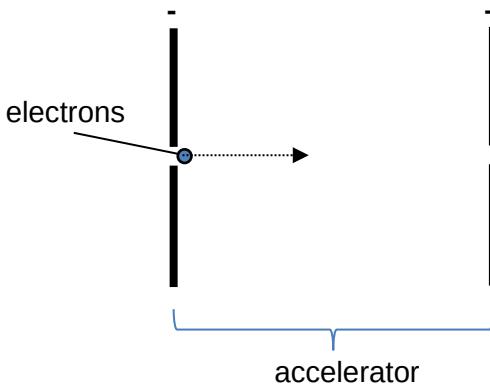


8	A mass spectrometer separates charge particles based on mass-to-charge ratio so that the composition of the charge particles can be identified. The schematic diagram of a type of mass spectrometer is shown in Fig. 8.1. There are three sections to this mass spectrometer – the accelerator, the velocity selector and the mass separator.  Fig. 8.1 Electrons are ejected into the mass spectrometer to demonstrate the working principle of the mass spectrometer.
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	(a)	(i) Electrons enter the mass spectrometer at the accelerator near to the negatively charged plate so that they accelerate towards the positively charged plate as shown in Fig. 8.2.  Fig. 8.2 The kinetic energy of the electrons increases by $2.50 \times 10^{-16} \text{ J}$ between leaving the negatively charged plate and reaching the positively charged plate. Calculate the accelerating potential difference (p.d.).
		accelerating p.d. =V [2]

			[3]
L2	The electrons will move in a <u>parabolic path</u> .		B1
	Electrons in a uniform electric field will experience a constant electric force upwards towards the positively charged plates throughout its motion . Therefore, the electrons will accelerate uniformly upwards .		B1
	As there is no horizontal force on the electrons, the horizontal velocity will remain unchanged .		B1
(c)	A uniform magnetic field is subsequently applied to the region in between the parallel charged plates such that only electrons with specific velocity pass through the velocity selector undeflected .		
	(i) State the direction of the magnetic field.		
			[1]
L1	(Perpendicular and) <u>into the page</u> .		B1
	(ii) Calculate the magnetic flux density in the velocity selector if the electrons that are undeflected have a speed of $3.25 \times 10^6 \text{ m s}^{-1}$ after passing through the fields.		
			
	magnetic flux density = T		[3]
L2	The electric field in between the two parallel plates, $E = \frac{\Delta V}{d} = \frac{1500}{16 \times 10^{-3}} = 93750 \text{ N C}^{-1}$ By Newton's second law, as the magnetic force is equal in magnitude to the electric force on the undeflected electrons, $Bqv = qE$ $B = \frac{E}{v} = \frac{93750}{3.25 \times 10^6} = 0.028846$ $B = 2.88 \times 10^{-2} \text{ T}$		C1 M1 A1
(d)	At the mass separator, the electrons then enter a region of uniform magnetic field set up by a large solenoid. The solenoid has 120 turns for every 15 cm of the solenoid. The current in the solenoid is 3.5 A.		
	(i) Calculate the magnitude of the magnetic flux density B at the centre of the solenoid due to the current of 3.5 A.		
			
	$B = \dots\dots\dots \text{ T}$		[2]
L1	The magnetic flux density at the centre of the solenoid, $B = \mu_0 n I = 4\pi \times 10^{-7} \left(\frac{120}{15 \times 10^{-2}} \right) 3.5$ $B = 3.5186 \times 10^{-3} = 3.52 \times 10^{-3} \text{ T}$		M1 A1
	(ii) Inside the dashed region on Fig. 8.4, sketch the magnetic field pattern due to the current in the solenoid.		

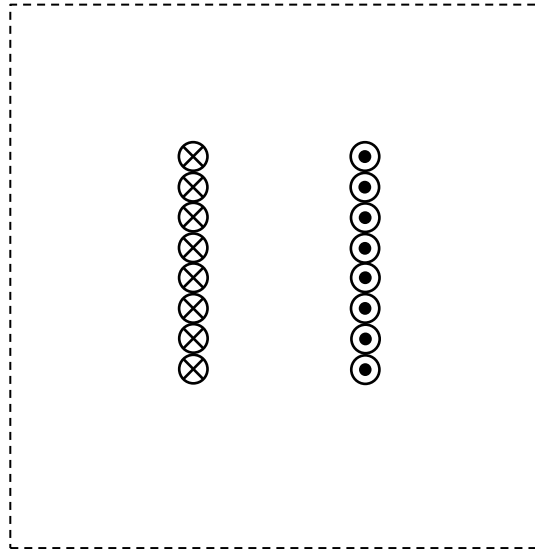
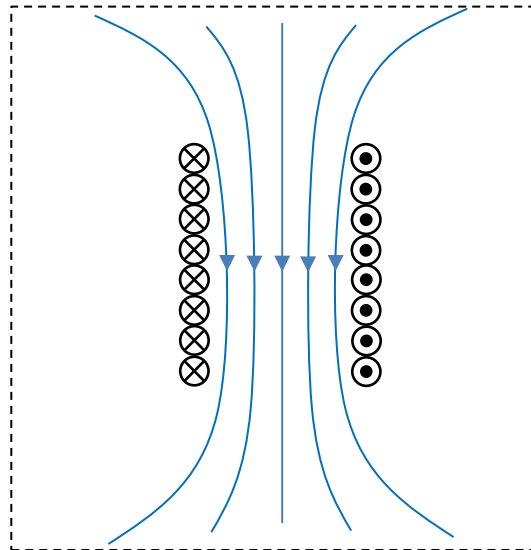


Fig. 8.4

[3]

L2

Solution:



1 mark – illustrate uniform strength of field through spacing between lines – equal spacing at the centre of the solenoid and

1 mark – illustrate weaker strength of field – large spacing at the ends

1 mark – direction of all field lines – arrows downwards

B1

B1
B1

(iii) The electrons enter the region of the uniform magnetic field perpendicularly.

Explain why the path of the electrons in the magnetic field is circular.

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				[3]
	L2		As the electrons are charged and enters the uniform magnetic field perpendicularly, there is a magnetic force that is always perpendicular to both the electrons' velocity and the field. As the magnetic force is always perpendicular to the electrons' velocity, the force continuously changes the electrons' velocity direction but not the velocity magnitude. The magnetic force will be constant at constant speed, and provides for the centripetal force of the electrons, causing the electrons to travel in a circular path of constant radius.	B1 B1 B1
		(iv)	In usual application, charged particles of different masses enter the mass separator instead of just electrons. Suggest how the uniform magnetic field can separate the charged particles by mass.	
				[2]
	L3		As the magnetic force provides for the centripetal force, $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ thus, when the charged particles of the same speed v (after passing through the velocity selector all the particles have same speed) move in the same uniform magnetic flux density B, the radius r of the circular path of the charged particles is proportional to the mass m but inversely proportional to the charge q (proportional to the mass-to-charge) ratio. Different charged particles have different mass-to-charge ratio, thus separated by moving in circular path with various respective radius.	B1 B1

[Total: 20]

